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SECOND SEMESTER

Roll No.....

B.E. (COE)

B.E. END SEM. EXAMINATION, May-2019

CEC02: Data Structures

Time: 3:00 Hrs.

Max. Marks: 40

Note: Attempt ALL questions. Each question carries equal marks

Assume suitable missing data, if any

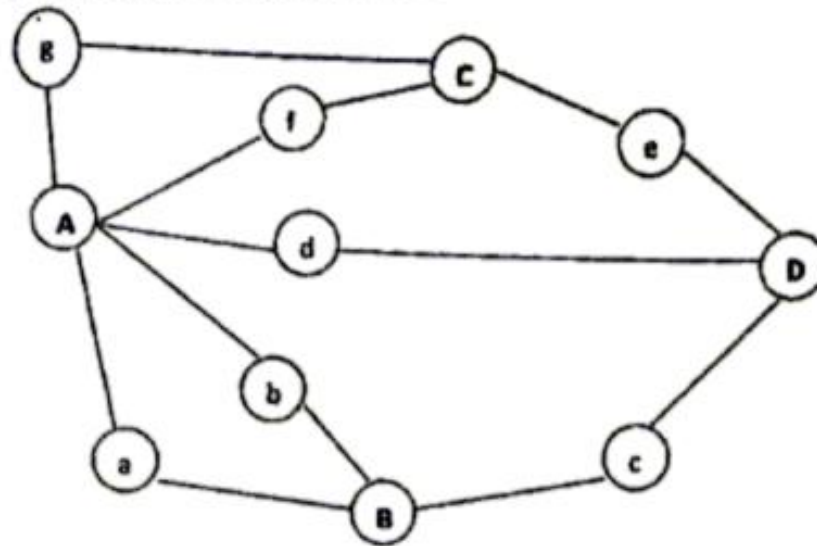
- 1) In each of the following question, please specify if the statement is true or false. If the statement is true, explain why it is true. If it is false, explain what the correct answer is and why? (5)
 - i) Linked lists are collections of data items "lined up in a row"—insertions and deletions can be made only at the front and the back of a linked list
 - ii) A null pointer normally indicates the end of a data structure.
 - iii) Binary trees facilitate high-speed searching and sorting of data, efficient elimination of duplicate data items, compilation of expressions into machine language and many other interesting applications
 - iv) The size of lists, stacks, queues and trees must be determined in advance of using these data structures to ensure that the proper amount of memory is allocated for their elements.
 - v) Compilers use stacks to evaluate arithmetic expressions and generate machine language code to process the expressions.
- 2) Define and explain the following (with example): (5)
 - i) Circular queue
 - ii) External sort
 - iii) Complete binary tree
 - iv) Linear search
 - v) Abstract data type
- 3) Answer the following questions: (5)
 - i) Write an algorithm for removing duplicate nodes in a given unsorted linked list?
 - ii) Differentiate between binary search and binary search tree. Give the situation where you would like to use binary search instead of binary search tree. Justify your
- 4) Answer **any TWO** of the following questions: (5)
 - i) A binary tree is threaded by making all right child pointers that would normally be null point to the inorder successor of the node (if it exists), and all left child pointers that would normally be null point to the inorder predecessor of the node. Write an algorithm to create threaded binary tree.
 - ii) Write an algorithm to find whether a given undirected graph is connected or not.
 - iii) Explain Trie Data Structure with example.
- 5) Answer the following questions: (5)
 - i) What is an AVL tree? Explain different rotations of AVL tree with example
 - ii) Sort the following using quick sort:
1, 12, 8, 2, 25, 6, 14, 28, 17, 7, 52, 16, 48, 68, 3, 26, 29, 53, 55, 45, 67
Show all the steps.

- 6) Write algorithms for **any TWO** of the following:
- To convert infix expression to postfix expression
 - To insert the node in doubly linked list
 - To create an expression tree.

(5)

- 7) Answer the following questions:

- Explain Breadth First Search and Depth First Search algorithms.
- Consider the following graph. In what order will the nodes be visited using a Breadth First Search and Depth First Search if start from d?



- 8) Write short notes on any TWO the following:

- Btree
- Huffman Coding
- Topological sort
- Merge sort

(5)